EXSOIL NSF PROTOTYPE · PHASE OBJECTIVES

Scientific capability on the working container

The phase charter: with a working multi-arch CTSM container in hand, deliver the three analysis Hubs on live data, validated across five NEON sites.

Data Science Institute, UW-Madison · July 16, 2026

PRIOR PHASE

Container ✅

THIS PHASE

3 Hubs live

SITES

5 NEON towers

DELIVERABLE

Local Docker MVP

Where we are

The infrastructure phase is complete. The project has a self-maintained, multi-architecture (amd64 + arm64) container running **standalone CTSM 5.4.043**, and a NEON tower simulation runs end-to-end to completion (KONZ baseline: 83 monthly history files, full 2018-2024 transient run).

> **This document marks the phase boundary:** infrastructure (done) → scientific capability (this phase). It states what the phase is trying to achieve and how we will know it succeeded. Execution detail lives in the implementation plan and GitHub epic #11; this is the charter above them.

Objective

Deliver the three analysis **Hubs** running against **live/native data** on the working container, validated across **5 NEON sites**. Each Hub realizes one capability from Jingyi's NSF proposal:

HUB	CAPABILITY	WHAT IT DOES
1 — Data Analysis	Data access	Explore NEON + CTSM data (plots, histograms, soil profiles)
2 — Modeling	Model evaluation	Evaluate fit and calibrate via a Kalman filter (target: **GPP**)
3 — Experimentation	Scenario analysis	Perturb an input, run CTSM on perturbed + unperturbed, compare via t-test

The Hubs, their analytics backend, and the perturbation tooling already exist; this phase repoints them from the original S3 fixtures to live/native data and validates them. It is a **data-rebind + validation** effort, not a redesign.

Sample sites (5): KONZ (baseline), ABBY, CPER, TALL, CLBJ.

Success criteria

This phase is done when:

1. All three Hubs run **end-to-end on live/in-container data** with no manual data-staging steps.
2. Each Hub runs **clean across all 5 sample sites** from a fresh container pull.
3. Hub outputs **validate against the reference copies** within an agreed tolerance (shape / plausible-range, not bit-level — see constraints).
4. Hub 2's Kalman/misfit step evaluates model **GPP against observed GPP** (pending an observation source).
5. The workflow is documented so a new user can run all three Hubs from the getting-started guide.

Scope

In scope (MVP)

- Local Docker image, **single-user, no authentication**
- The three Hubs on the 5 sample sites, live/native data
- Precipitation perturbation for Hub 3 (fully codified today)

Out of scope (follow-on)

- VM hosting and multi-user access
- User/password authentication and S3 credential-rotation security work
- PFT / soil-type / temperature perturbations (unless promoted by Jingyi)

Open decisions (need Jingyi)

DECISION	STATUS
Perturbation variable scope — precip-only for MVP, with PFT / soil / temperature as follow-ons?	OPEN
Validation tolerance — how close must a live run match the references to pass? (shape/range framing)	OPEN

Dependencies & known constraints

ITEM	IMPLICATION
Observed GPP for Hub 2	Reference copies are **model output only** — no NEON observations. Hub 2's misfit needs observed GPP, sourced separately (NEON API / `evaluation_files` / NCAR eval files). Main external dependency.

ITEM	IMPLICATION
Reference copies = older model generation	Legacy `h1` / `h0` naming, 2018-2022 coverage. A shape / plausible-range oracle, not exact ground truth. Confirm version with Maria.
Model vs observation not exact by construction	Model GPP is "gross"; tower GPP is derived from net flux via partitioning. Even a perfect model won't match exactly — fit-quality over exact agreement.

Scope discrepancies to reconcile

While mapping `communication-internal` against the code, three features are described (two marked "done") that do not match what is actually implemented. These are **scope decisions for Jingyi**, not silent omissions — flagged so they are not later read as delivered:

FEATURE (PER DOC)	ACTUAL STATE	DECISION / ACTION
ILAMB metrics MARKED DONE	No ILAMB in code; Hub 2 uses a custom fit framework (bias, R^2, residuals)	Accept custom or build real ILAMB — #13
5-step EnKF loop MARKED DONE	Code has a simple/scalar Kalman filter; no ensemble or cycle	Accept simple KF or build true EnKF — #14
Perturbation: PFT, soil, temperature	Only precipitation is codified today	MVP = precip; rest deferred (#3 soil type)
Observed data for evaluation	Reference copies are model-output only	Source observed GPP — #12 (blocks Phase 3)

The three-hub integration itself (Phases 0-5) is concretely planned and tracked; these four items are the remaining gaps between the source doc and the plan.

Related documents

- **Hub integration implementation plan** — `docs/hub-integration-plan/` — phases, tasks, acceptance criteria
- **Data contract** — `docs/data-contract.md` — live output schema + reference-copy evaluation
- **GitHub epic #11** — execution tracker (phase issues #5–#10)
- **`docs/communication-internal.docx`** — the July 9 decisions that set this phase's direction